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# SONIC-K1: The First Open-Weights Universal Motion Tracker for the Booster K1 Humanoid

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## 01 The idea: SONIC for a small humanoid

NVIDIA SONIC showed that a single policy can track almost any human motion, but it was built for the larger Unitree G1 and trained on a large GPU cluster. We asked whether a lab with one GPU can have the same kind of tracker on a small robot. So we ported SONIC to the Booster K1, trained it from scratch on one GPU, fixed it for the real robot, and released everything.

| | SONIC · Unitree G1 as reported | SONIC-K1 · Booster K1 ours |
|---|---|---|
| Held-out success | 98.5% | **97.4%** |
| PHUMA success | 97.2% | **93.6%** |
| Training data | 700 h mocap | **9,991 clips** |

G1 values as reported by SONIC; robot, data and test clips differ.

### What we contribute

- The first open-weights general motion tracker for the Booster K1, to our knowledge.
- A full SONIC training run on one GPU, under 1% of SONIC's reported compute.
- Tested in Isaac Lab, in MuJoCo and on the real K1, with a short post-training for the robot's real torque limits and delay.
- Code, weights, robot app and evaluation tools, all public.

### Training time, drawn to scale

SONIC — 128 GPUs × 7 days = 21,000 GPU-h

SONIC-K1 — 1 GPU = ~145 GPU-h

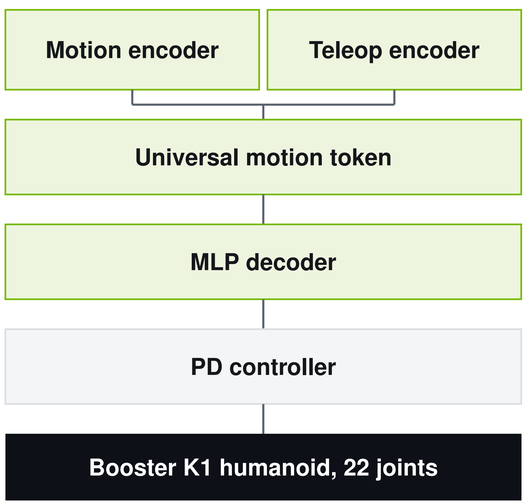
135.4 h of base training on one NVIDIA H100, plus 9.2 h of post-training on one RTX PRO 6000.

## 02 What we built: SONIC on the K1

SONIC is one network that tracks any reference motion. We kept its design, PPO settings, motion sampling and rewards, trained both encoders from scratch on K1 motion, and added a head-joint reward.

We retargeted 142,220 Bones-SEED human clips to the K1 with UMR and kept the 62,427 that the robot can physically perform, training on part of them and holding out 1,985 for testing. 16,384 simulated K1s then learn in parallel in Isaac Lab with GPU PhysX, all on one NVIDIA H100. A short post-training on one RTX PRO 6000 adds the robot's real torque limits and a random delay, and the final policy is exported to ONNX and runs onboard the K1 at 50 Hz. Simulation and training run end to end on the NVIDIA stack.

### The SONIC network on the K1




Green = NVIDIA tools (the SONIC network)

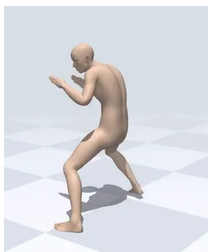
**Post-trained:** a short extra run with the K1's real torque limits and a random delay.

### Built with NVIDIA

- **NVIDIA SONIC** network and recipe
- **GR00T-WBC** training code
- **Isaac Lab** RL training
- **Isaac Sim + PhysX** GPU simulation
- **NVIDIA H100** base training
- **RTX PRO 6000** post-training
- **SOMA** human body model

## 03 Tested in simulation and on the real K1

Every test starts from a human motion the policy has never seen, stored as a SOMA body (left). We retarget it to the K1 and play it in MuJoCo, a different physics engine from training, and then on the real robot (right).



**Human reference,** SOMA body



**SONIC-K1,** same motion

**99.0%** of 100STYLE motions tracked (external set)

**96.6%** from 76.7% post-trained, torque limits + 40 ms delay

A clip counts if root, hands and feet stay within 0.25 m of the reference height (SONIC's rule).

| 98.7% | 97.1% | 91.7% | 1.1% |
|---|---|---|---|
| stay upright in Isaac Lab | held-out, robot limits + 20 ms delay | PHUMA, robot limits + 40 ms delay | steps at a torque cap on the real K1 |

### Conclusion

A universal humanoid motion tracker can be ported to a small humanoid on a single GPU. To our knowledge, SONIC-K1 is the only public general motion tracker for the Booster K1: it tracks almost every held-out motion at a small fraction of SONIC's compute and runs on the real robot.

**Future work:** tracking error on hardware, teleoperation, video input.

### Everything is open

Code, robot app, eval tools · Apache-2.0 · github.com/Kyk0/sonic-k1
Weights v1.0 + v0.1, ONNX · CC BY-NC 4.0 · huggingface.co/Kyry10/sonic-k1
K1 USD asset, test-split list, laptop demo · kyry.me/sonic-k1

### Limits and related work

Mostly simulation, one training seed, post-training changes not ablated. Any2Any and TRACE transfer G1 SONIC to full-size robots; we train from scratch for a smaller one.

### Thanks

NVIDIA GEAR, Bones Studio, the UMR authors and Booster Robotics. Training data includes Motion Data by Bones Studio.